

Air Quality Health Index (AQHI) — Edmonton

3.0

Current AQHI at Edmonton Lendrum (18.14 km from venue)
Forecast Sun 4:00 PM at Edmonton Lendrum (18.14 km from venue): **3.5**

Health Risk	AQHI	At-Risk Population	General Population
Low Risk	3.5	Enjoy your usual outdoor activities.	Ideal air quality for outdoor activities.

No active Environment Canada weather alerts for the venue.

Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 24.6°C and feels near 27.5°C. Winds are 5 km/h from N, gusting near 17 km/h.
- The nearest current AQHI is 3 at Edmonton Lendrum, 18.1 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 5 (forecast tonight: 4, tomorrow: 2).
- A gridded AQHI estimate blending official and community sensors (confidence: High, nearest contributing point 6.1 km away) puts the area near 4.
- The nearest air monitoring station (Leduc Sensor, 12.7 km) reports fine particulate matter (PM2.5) at 38.7 µg/m³.
- A nearby community sensor ('ACA_RMA', 7.4 km) reads 45.9 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 37.6 km to the NW, i.e. arriving from the NW.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-08-22) is 20.8 km N of the venue. The modeled back-trajectory passes within 19.2 km of this fire, making smoke transport to the venue plausible based on wind-field modeling.
- The AQHI forecast for Sun 4:00 PM is 4, based on Edmonton Lendrum, 18.1 km from the venue.
- Thunderstorm conditions appear in the hourly forecast beginning around Sun 8:00 PM.
- Overall operational environmental risk is HIGH. Primary concerns are heat, precipitation, thunderstorm.

Recommendations

- Confirm lightning monitoring, shelter, pause and evacuation procedures.
- Increase hydration, shade, cooling and heat-illness messaging.
- Prepare drainage, electrical protection and wet-weather controls.
- Environment Canada AQHI guidance — general population: Ideal air quality for outdoor activities.
- Environment Canada AQHI guidance — at-risk populations: Enjoy your usual outdoor activities.

Hazard Snapshot — Overall Risk: High Risk

AIR QUALITY Low Risk 3.5 AQHI	HEAT High Risk 30.0 °C apparent/humidex	WIND Low Risk 37.4 km/h peak gust	PRECIPITATION High Risk 3.6 mm/h maximum	THUNDERSTORM High Risk Sun 8:00 PM first forecast signal
AQHI RATE OF CHANGE Low Risk 0.0 AQHI change/hour		WIND SHEAR Low Risk 5.7 ° direction diff (10m vs 120m)		

Local Monitoring

Nearest station 3.0 Edmonton Lendrum · 18.14 km	Blended estimate 4.0 confidence High	PM2.5 (station) 38.7 µg/m³ Leduc Sensor · 12.73 km	PM2.5 (community) 45.9 µg/m³ ACA_RMA · 7.4 km
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